

ROBO RUMBLE 2026 · ROBO GRAND PRIX

Lightning McQueen

An autonomous line-following racer that picks its speed from the corner it is about to enter — not the one it is already in.

Team CO Coders

Elimination Round submission

0.86 kg

against a 5 kg limit

300 × 211 mm

against 500 × 500 mm

R 1 843.60

total cost of implementation

The category sets the problem. The constraints set the design.

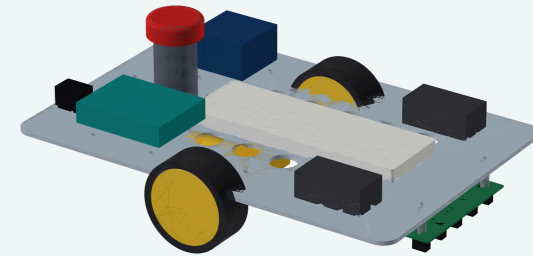
Robo Grand Prix is autonomous racing, so there is no problem to validate — the problem is given: get a robot around a line-marked circuit, on its own, faster than everyone else.

What there **is** to address is a set of hard constraints. A build either satisfies them or it does not.

- **Footprint $\leq 50 \times 50$ cm.** We are at $300 \times 211 \times 107$ mm, and the CAD build script asserts it on every run.
- **Mass below 5 kg.** We are at 0.86 kg — 17 % of the allowance.
- **Accessible ON/OFF switch, and an Emergency Stop.** Both drawn, annotated and placed where an operator can actually reach them.
- **Fully autonomous.** No radio, no Wi-Fi, no Bluetooth anywhere on the vehicle. Any human intervention forfeits the run.

Assembled chassis — isometric

300 × 200 mm acrylic deck · 170 mm track · 65 mm drive wheels · overall envelope 300 × 211 × 107 mm against a 500 × 500 mm limit



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What we are building

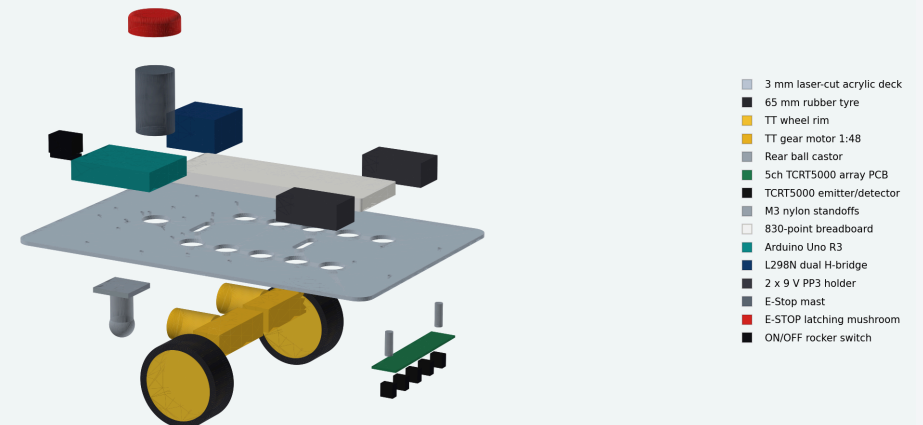
- **Sensing.** Five-channel TCRT5000 reflective array, 20 mm pitch, 12 mm above the floor, mounted 138 mm ahead of the drive axle.
- **Fusion.** A weighted centroid over five per-channel-normalised analog readings gives a cross-track error **in metres** — not a unitless index.
- **Control.** A PID whose output is a **path curvature** in 1/m, running at 200 Hz on an Arduino Uno.
- **Drive.** Two TT gear motors through an L298N H-bridge, from two 9 V PP3 cells on a single switched rail.
- **Structure.** One 3 mm laser-cut acrylic deck, 300 × 200 mm. Every joint bolted; nothing glued.

BUILDABLE, TODAY

Every part is stocked by South African hobby suppliers. The deck is one DXF and ninety seconds on a laser cutter. Nothing here waits on an import order.

Exploded assembly — fastening and stack-up

Deck modules bolt down on M3 nylon standoffs · motors, castor and sensor boom hang below the deck · every joint bolted, none glued



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Steer in curvature. Choose speed from the corner you are about to enter.

The naive line follower adds its error to one motor's PWM and subtracts it from the other. The gain that feels right at low speed oversteers at high speed — so you end up tuning one set of gains per speed.

Our PID instead outputs a **path curvature** κ . A demand of $\kappa = 5 \text{ 1/m}$ is a 200 mm radius turn at any speed. One set of gains, every speed.

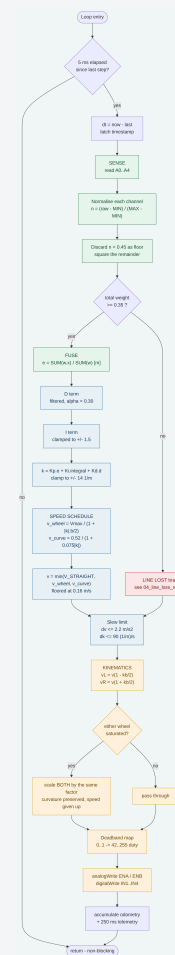
- **Speed comes from curvature, before the curvature is applied.** Two ceilings — the point where the outer wheel saturates, and a grip margin — and we take the lower.
- **The sensor bar leads the axle by 138 mm.** That is the distance in which the car sheds speed, so it arrives at the corner already slowed.
- **On saturation, both wheels scale together.** The path curvature survives; only the speed is given up. Clamping one wheel changes the radius and runs you wide.

200 Hz

control loop, 15 % CPU

5

analog channels — a continuous error, so the derivative term is usable



The E-Stop is the first device in the battery line

- **SW1 — E-Stop.** 22 mm latching mushroom head, normally closed, on a 48 mm mast at the rear centreline. Tallest object on the car, nothing overhanging it.
- **SW2 — ON/OFF.** SPST rocker at the rear-left corner, operable with the car on the grid — no lifting, no reaching over a wheel.
- **Everything is downstream.** The L298N's 78M05 derives the 5 V logic rail from the same switched line, so one strike de-energises the motors, the H-bridge, the MCU and the sensors together.
- **Firmware cannot defeat it.** The contact is mechanically in series with the battery. Nothing is left running once it opens.
- **Stored energy cannot move the car.** 100 μF at 5 V is 1.25 mJ; turning a loaded TT gearbox needs about a joule.

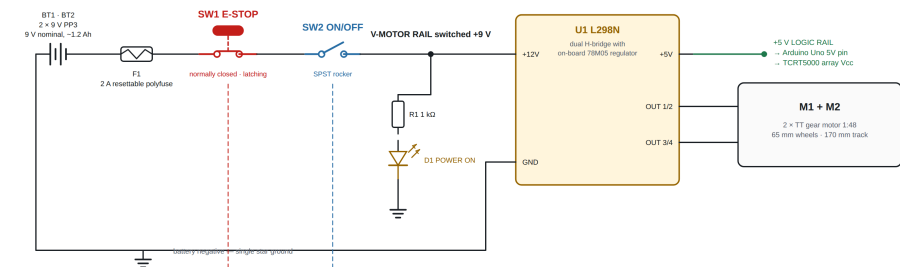
WHY THERE IS NO SEPARATE LOGIC BATTERY

A second supply would keep the MCU alive after the motor rail opened — exactly what Constraint 4 exists to prevent. One rail, one stop.

Power and safety chain

The mandatory ON/OFF switch and Emergency Stop, and why they sit where they do in the circuit

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SW1 - E-STOP — Universal Design Constraint 4
22 mm latching mushroom head, normally closed, on a 48 mm mast at the rear centreline of the deck. Nothing overhangs it.
It is the FIRST device in the battery positive line, upstream of SW2 and of U1. Sinking it removes power from the motors, the H-bridge, the 5 V regulator, the MCU and the sensor array in the same instant. It latches down, so power cannot restore itself until an operator twists the head back out. The contact is mechanically in the battery line, so no firmware path can defeat it.

SW2 - ON/OFF — Universal Design Constraint 3
SPST rocker at the rear-left corner of the deck, clear of the wheels and of the sensor boom.
Operable with the car sitting on the grid — no lifting, no tilting, no reaching over a wheel that is about to spin.
It is the normal way to power the car up and down. SW1 is reserved for stopping a run that has gone wrong.

Downstream of both devices
U1 derives the +5 V logic rail from the same switched line, so nothing keeps the MCU alive after a stop. The largest capacitor downstream is 100 μF on the L298N rail — far too little stored energy to turn a loaded TT gearbox even once.

05 · COST OF IMPLEMENTATION

R 1 843.60, all in

CATEGORY	COST	SHARE
Control & drive electronics	R 480.00	28.6 %
Chassis & hardware	R 440.00	26.3 %
Power & safety	R 301.00	18.0 %
Interconnect	R 230.00	13.7 %
Sensing	R 167.00	10.0 %
Indication & passives	R 58.00	3.5 %
Contingency — spares and breakage (10 %)	R 167.60	—

TOTAL COST OF IMPLEMENTATION R 1 843.60

24 line items, 28 units. Full breakdown with quantities, suppliers, stock codes and buy links in [Bill_of_Materials.xlsx](#).

0.86 kg

assembled mass — 17 % of the 5 kg allowance. Two 9 V PP3 cells weigh 92 g where twelve AA cells weigh 276 g.

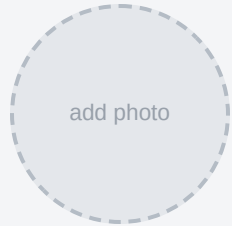
R 145

the E-Stop — the most expensive single item, and the one line we will not economise on.

R 180

laser cutting the deck. Buys dimensional accuracy we cannot get by hand, and a re-cut takes ninety seconds.

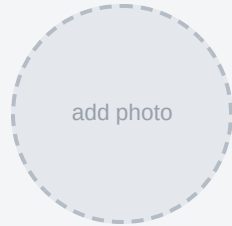
Team CO Coders



[Name]

Team lead · Controls

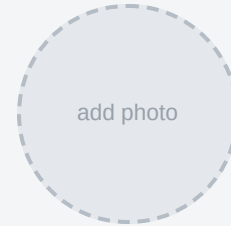
Control law, firmware architecture, Webots simulation and the speed schedule.



[Name]

Electronics

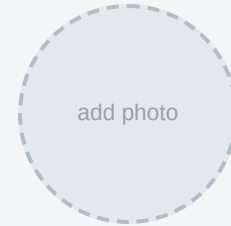
Component selection, schematics, power and safety chain, wiring loom.



[Name]

Mechanical

Parametric CAD, laser-cut deck, fastening scheme, mass budget.



[Name]

Integration · Documentation

Assembly, calibration, bill of materials, repository and submission.

REPLACE THE FOUR PLACEHOLDERS BEFORE SUBMITTING

Add each member's photo and name. The Team Slide is a scored requirement: photos of members and their roles.